

1024 x 32 (4 KB) SKY130 SRAM

Post-Layout Datasheet - OpenRAM / SKY130 - 1.8 V - Single-Port 1RW

Design: task5_4kb_sram	Logical Capacity: 32,768 bits / 4 KB	Status: POST-LAYOUT VERIFIED
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1. Post-Layout Physical Specifications

Parameter	Post-layout result
Technology	SKY130
Memory organization	1024 words x 32 bits
Logical capacity	32,768 bits / 4 KB
Port configuration	Single-port 1RW
Logical address width	10 bits
Logical data width	32 bits
Supply voltage	1.8 V
OpenRAM version	1.2.49
Words per row	8
Macro width	808.845 um
Macro height	351.29 um
Macro area	284,139.16 um^2
Extraction basis	Capacitance-extracted (C-extracted)

2. Physical Verification Summary

Check	Result	Evidence / interpretation
Functional regression	PASS	144 PASS / 0 FAIL
Magic DRC	PASS	0 violations
Netgen LVS	PASS	Unique circuit match; 270248/270248 devices; 69617/69617 nets; 0 property errors
Extraction	PASS	C-extracted hierarchy used for targeted post-layout characterization
RTL integration	PASS	13 PASS / 0 FAIL
OpenSTA timing	PASS	Positive setup and hold slack at the reported SS timing-model point

3. Logical Interface

Signal	Width	Function
clk0	1	Clock
csb0	1	Active-low chip select
web0	1	Active-low write enable
addr0	10	Logical word address
din0	32	Write data
dout0	32	Read data

Post-Layout Characterization

Targeted C-extracted results at 1.8 V across TT / SS / FF

4. Extracted Timing, Power and Energy

Parameter	TT	SS	FF	Unit
Worst write time	77.9407	117.096	60.6197	ps
Read access: SEN50 to DOUT50	40.8805	71.8089	N/A*	ps
Leakage power	3.8723	7.0853	8.2157	uW
Read energy	31.3736	42.3984	42.0730	fJ
Write energy	95.0453	177.7866	101.2986	fJ

* FF read event ordering: DOUT reaches the 50% point before the SEN50 reference, so a SEN50-to-DOUT50 delay is not reported as a fabricated zero. The measured FF WL50-to-DOUT50 value is 392.222 ps.

5. Timing-Model Sign-Off

Parameter	Value
Worst characterized corner	SS
Minimum characterized period	2.241 ns
Liberty-model frequency boundary	446.229362 MHz
OpenSTA setup slack	+2.138 ns
OpenSTA hold slack	+0.056 ns
OpenSTA result	PASS

6. Reporting Notes

- **Post-layout only:** This datasheet excludes preliminary Task 4 results, pre-layout values, and reference-repository analytical values.
- **Extraction scope:** Characterization is capacitance-extracted. It must not be described as RC-extracted unless extracted resistor evidence is separately established.
- **Frequency interpretation:** 446.229362 MHz is the characterized Liberty-model boundary for the reported SS minimum period; it is not a guard-banded system operating frequency.
- **Interface note:** The physical OpenRAM macro includes one spare row and one spare column; the logical wrapper preserves the required 1024 x 32 user-visible interface.

Final post-layout status	READY FOR DATASHEET SUBMISSION
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Prepared from the final Week 6 Task 5 verification and characterization evidence for the 4 KB SRAM design.